Ahmed Hannan

Full-Stack Software Engineer & AI Researcher

ahmedhanan4@gmail.com • +92 333 149 5251 • linkedin.com/in/ahmedhannan2 • github.com/ahmed-hannan • Islamabad, Pakistan

Professional Summary

Full-Stack Software Engineer and AI Researcher with hands-on production experience architecting scalable web and mobile applications (iOS/Android) and deploying end-to-end deep learning pipelines. Proven expertise in Generative AI and Multimodal RAG (CLIP, FAISS, Gemma 3), cross-encoder semantic search (BERT, Amazon ESCI), real-time computer vision (YOLO, Unity VR), GPU compute acceleration (CUDA, Tensor Cores), and modern full-stack frameworks (React, Next.js, Node.js, Django, Flutter). Combines rigorous AI research experience at the National Center of AI (NCAI) with production engineering at EmpowerBits (awarded Performance Certificate) architecting high-concurrency client solutions with sub-second latency and resilient microservices.

Education

National University of Computer & Emerging Sciences (FAST-NUCES), BS in Computer Science — Graduation: June 2026

- Relevant Coursework: Distributed Systems, High-Performance Computing, Deep Learning, Generative AI, Machine Learning, Computer Vision, Advanced Operating Systems, Database Systems, Software Engineering, Networks, Algorithms, Linear Algebra.

Technical Skills

Core AI & Deep Learning: PyTorch, TensorFlow/Keras, HuggingFace Transformers, Sentence-Transformers, Scikit-Learn
Generative AI & LLMs: Multimodal RAG, Prompt Engineering (Zero/Few-Shot, CoT), Fine-Tuning (BERT, Gemma), FAISS, spaCy, NLTK
Computer Vision & Edge AI: CLIP, YOLO (v8/v11), OpenCV, Real-Time Pose Estimation, Image Segmentation, Unity VR
HPC & GPU Optimization: CUDA, Tensor Cores, GPU Parallelization, OpenMP, MPI, Layer Offloading, FP16 Quantization
Languages & Core Tech: Python, C/C++, TypeScript, JavaScript, Dart, SQL, HTML5/CSS3, Bash
Frontend & Cross-Platform: React.js, Next.js, Flutter (iOS & Android), Tailwind CSS, Redux, State Management, Vite
Backend & Microservices: Node.js, Express.js, Django, FastAPI, Flask, RESTful APIs, WebSockets, Redis
Databases & Cloud: PostgreSQL, MongoDB, Firebase, Google BigQuery, Cloud SQL, SQLite
DevOps & MLOps Tools: Linux, Git/GitHub, Docker, CI/CD Pipelines, Jira, Agile/Scrum, Selenium, Apache Airflow

Professional Experience

EmpowerBits, Full-Stack Developer — 2026/06 – Present

- Engineered and deployed production-ready web and mobile applications (iOS & Android) actively serving concurrent user traffic.
- Built modular, reactive UI architectures with Flutter and React, optimizing state management to reduce client-side re-renders and latency.
- Designed resilient backend REST APIs and microservices, implementing database query indexing and connection pooling in PostgreSQL.
- Architected low-latency API services and reactive state management, reducing data fetch times by 30% under heavy concurrency.
- Awarded Certificate of Excellence / Performance Certificate for outstanding software delivery, technical ownership, and rapid release velocity.

RedCoast, Full-Stack Software Engineer — 2025/06 – 2026/06

- Spearheaded full-stack feature implementation and performance optimization across multi-tier client web and mobile applications.
- Managed one-to-one client interactions to translate operational specifications into scalable, production-ready architectural designs.
- Automated integration and regression testing using Selenium and Jira, significantly improving engineering velocity and release stability.
- Conducted large-scale query optimization and analytics on user engagement telemetry via Google BigQuery to optimize system bottlenecks.

National Center of AI (NCAI, NUST), AI Research Intern — 2024/08 – 2024/09

- Researched generative transformer architectures and multimodal image-text synthesis; authored formal technical documentation for hybrid AI deployments.
- Investigated deep learning architectures with a focus on computational efficiency, resource profiling, and technical documentation.

FAST – NUCES, Teaching Assistant (Programming Fundamentals) — 2024/06 – 2024/09

- Assisted in delivering core programming coursework; mentored 50+ students in C/C++ memory management, dynamic allocation, pointer arithmetic, and algorithmic problem-solving.

Featured Engineering & Research Projects

Pose2Play — AI-Powered VR Rehabilitation Platform (Final Year Project) GitHub: Pose2Play

- Built a real-time computer vision system using YOLO-based pose estimation to track 3D patient movement kinematics and deliver corrective feedback.
- Integrated low-latency Python model inference with Unity VR, achieving 30+ FPS real-time responsiveness to boost physical therapy adherence.
- Developed real-time inter-process communication (IPC) between Python kinematic estimators and Unity C# VR rendering components.

Multimodal RAG & Semantic Product Search Suite GitHub: multimodal-rag-semantic-search

- Engineered an end-to-end multimodal RAG pipeline for complex financial PDFs using PyMuPDF, OCR, and dual encoders (CLIP-512 + MiniLM-384) in FAISS.
- Deployed quantized Gemma-3-1b-it inference with FP16 and layer offloading for 3GB VRAM; fine-tuned BERT-Tiny cross-encoder on Amazon ESCI (1.1M records).

CRAG: Advanced Multi-Pipeline RAG Benchmark & Web Dashboard GitHub: crag-advanced-rag-pipelines

- Benchmarked 7 retrieval pipelines (RAG Fusion, HyDE, Corrective RAG with confidence gating, Graph RAG, RRR) on the Meta CRAG noisy web corpus.
- Developed a global FAISS vector index, spaCy NER answer extractor, and evaluation harness reporting normalized exact-match accuracy.
- Built a Flask REST API backend integrated with FAISS vector indexing and an interactive React + Vite frontend dashboard.

Parallel Implementation of ParButterfly on Large Bipartite Graphs HPC Graph Analytics

- Designed and implemented a scalable parallel algorithm for butterfly-motif counting in billion-edge bipartite graphs using OpenMP and MPI.
- Optimized cache locality, memory bandwidth, and load balancing across multi-core symmetric multiprocessing (SMP) and cluster environments.

High-Performance Deep Neural Network Acceleration GitHub: Neural-Network-Acceleration

- Accelerated deep neural network forward/backward passes and matrix kernels using CUDA, Tensor Cores, and GPU parallelization techniques.
- Profiled memory bandwidth bottlenecks, coalesced global memory transactions, and utilized shared memory tiling for maximum compute saturation.

GoMatch & Full-Stack Application Platforms GitHub: GoMatch, QuizVerse

- Architected a mobile ride-hailing app replicating Uber/Careem workflows with driver-rider matching, live geolocation tracking, and trip histories.
- Engineered responsive full-stack platforms featuring role-based access control, Google OAuth/JWT authentication, and real-time state management.

Honors & Leadership

- Certificate of Excellence / Performance Certificate: Awarded by EmpowerBits for high-impact production engineering and technical delivery.
- Academic Appointments: Appointed Undergraduate Teaching Assistant at FAST-NUCES for Programming Fundamentals.